



Advanced UI Development Techniques

Module 6 Chapter 1

Introduction

- Modern UI development must be **visually appealing, functional, scalable, and maintainable**.
- Applications should work smoothly across **multiple platforms and devices**.
- Advanced UI techniques help deliver **high-quality user experiences**.



Advanced Widget Usage

Use the full power of **built-in widgets**.

Create **custom widgets** for specific UI needs.

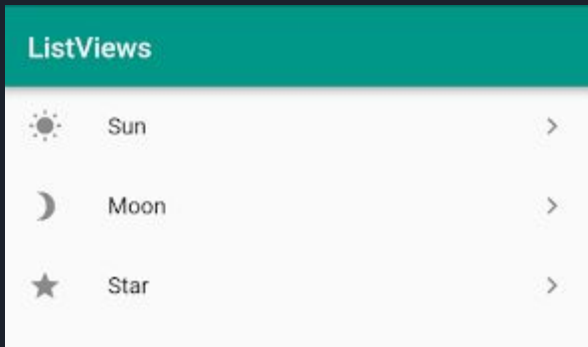
Leverage **widget libraries and frameworks** for faster development.



Built-in Widgets

ListView

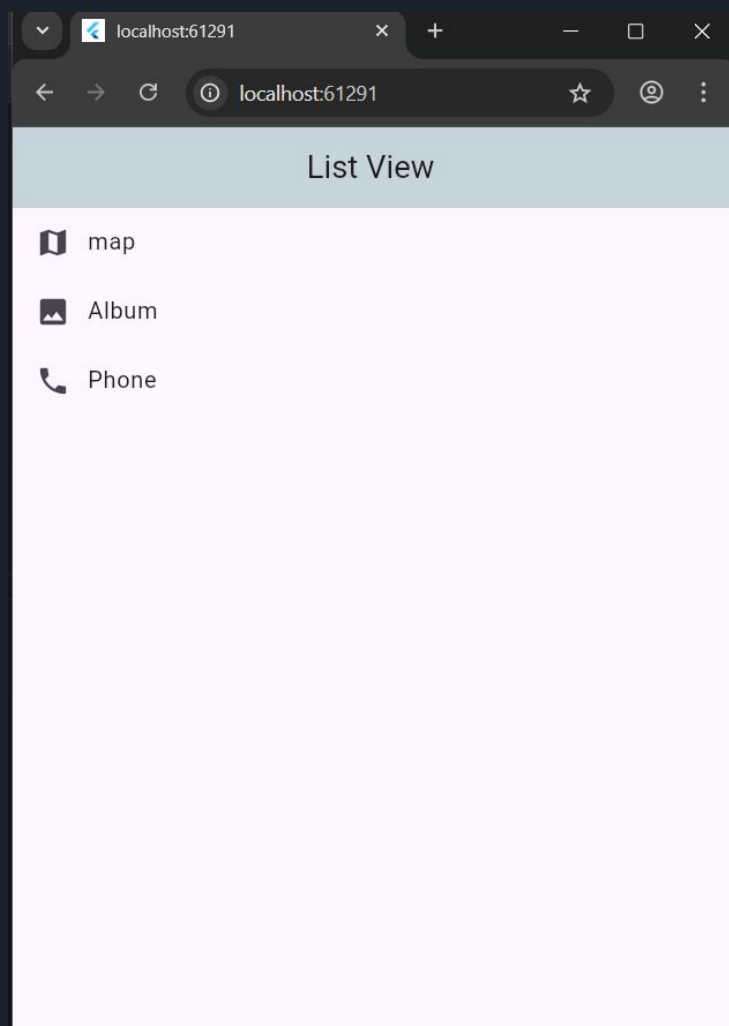
- Displays items in a **single direction** (vertical or horizontal).
- Commonly used for **text lists, images, and cards**.
- Types:
 - `ListView.builder` – for dynamic lists
 - `ListView.separated` – adds separators
 - `ListView.custom` – full layout control



```

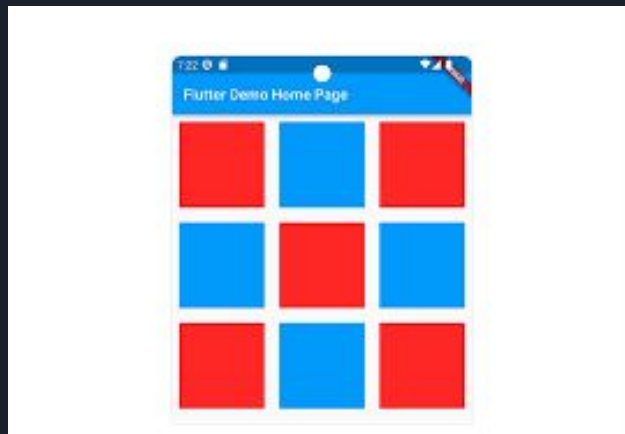
1 import 'package:flutter/material.dart';
2 class ListVieww extends StatelessWidget{
3   ListVieww({super.key});
4   @override
5   Widget build(BuildContext context) {
6     return Scaffold(
7       appBar: AppBar(
8         backgroundColor: const Color.fromARGB(255, 198, 213, 221),
9         title: Text('List View '),
10        centerTitle: true,
11      ), // AppBar
12      body: ListView(
13        children: [
14          ListTile(
15            leading: Icon(Icons.map),
16            title: Text('map'),
17            onTap: () {
18            },
19          ), // ListTile
20          ListTile(
21            leading: Icon(Icons.photo),
22            title: Text('Album'),
23            onTap: () {
24            },
25          ), // ListTile
26          ListTile(
27            leading: Icon(Icons.phone),
28            title: Text('Phone'),
29            onTap: () {
30            },
31          ) // ListTile
32        ],
33      ), // ListView
34    ); // Scaffold
35  }
36 }

```



GridView

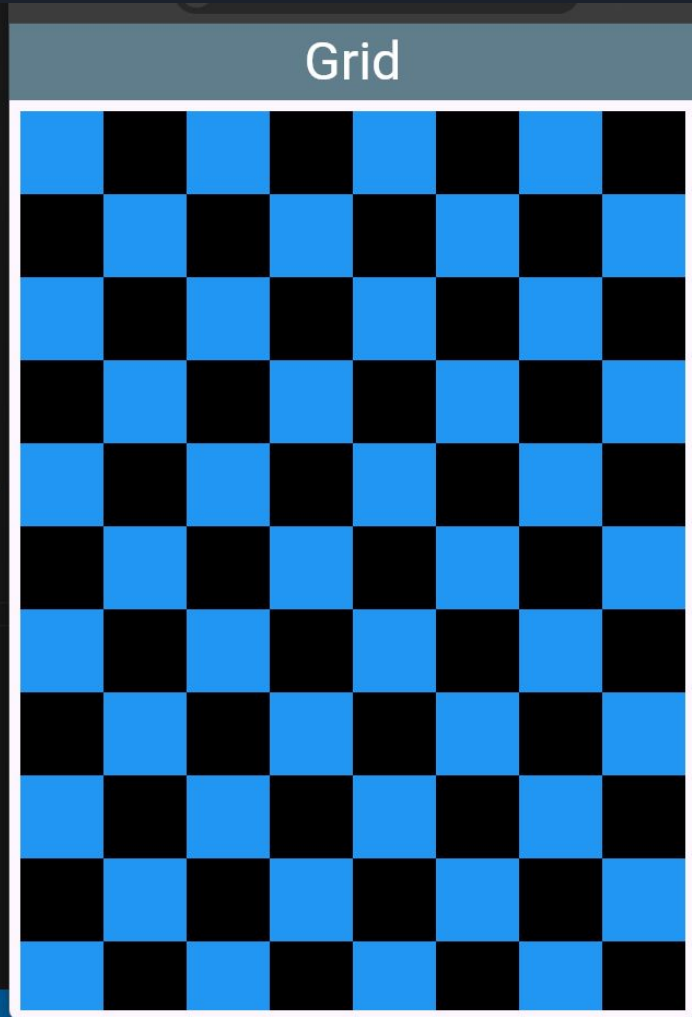
- Arranges items in **rows and columns**.
- Suitable for **photo galleries & product catalogs**.
- Types:
 - `GridView.count`
 - `GridView.builder`
 - `GridView.extent`



```

lib > grid.dart > _GriddState > build
1  import 'package:flutter/material.dart';
2  class Gridd extends StatefulWidget {
3    const Gridd({super.key});
4    @override
5    State<Gridd> createState() => _GriddState();
6  }
7  class _GriddState extends State<Gridd> {
8    @override
9    Widget build(BuildContext context) {
10     return Scaffold(
11       appBar: AppBar(
12         backgroundColor: Colors.blueGrey,
13         title: Text('Grid', style: TextStyle(fontSize: 39, color: Colors.white)),
14         centerTitle: true,
15       ), // AppBar
16       body: Padding(padding: EdgeInsetsGeometry.all(8),
17         child: GridView.builder(
18           gridDelegate: SliverGridDelegateWithFixedCrossAxisCount(crossAxisCount: 8),
19           itemBuilder: (context, index){
20             var xindex = index % 8;
21             var yindex = (index/8).floor();
22             return Container(
23               color: (xindex + yindex).isEven ? Colors.blue : Colors.black,
24               child: Stack(
25                 children: [
26
27                 ],
28               ), // Stack
29             ); // Container
30           }, // GridView.builder
31         ), // Padding
32       ); // Scaffold
33     }
34   }
35 }

```



Tab Bar

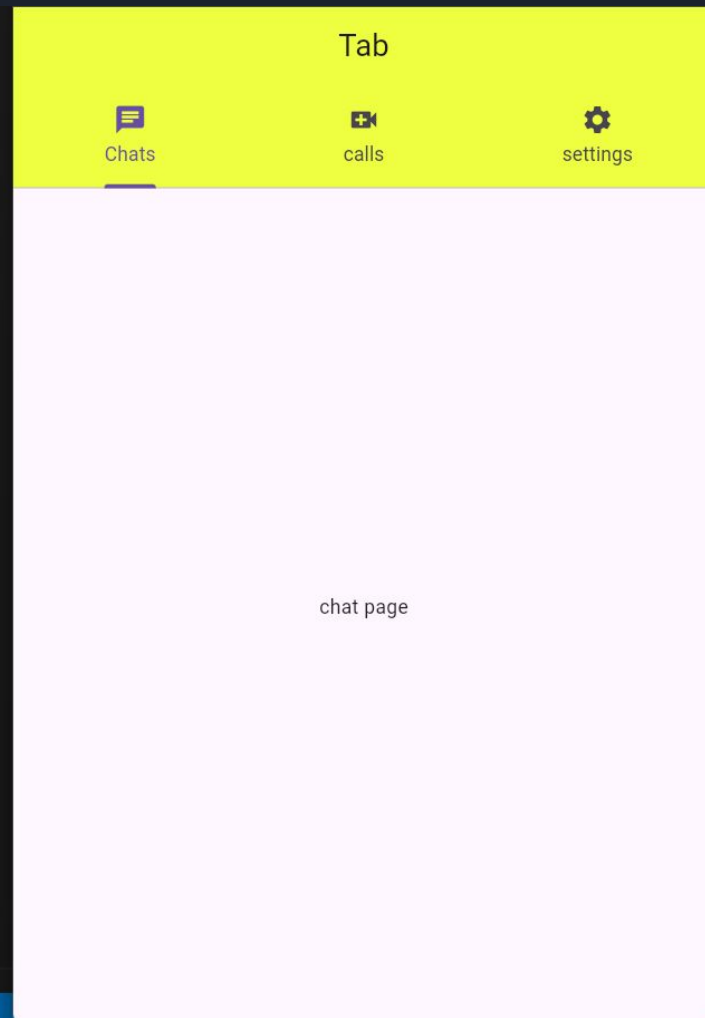
- Used to switch between different screens.
- Follows **Material Design** guidelines.
- Steps:
 - Create a TabController
 - Create tabs
 - Add content for each tab




```

1 import 'package:flutter/material.dart';
2 class Tabbs extends StatefulWidget {
3   const tabbs({super.key});
4   @override
5   State<tabbs> createState() => _tabbsState();
6 }
7 class _tabbsState extends State<tabbs> {
8   @override
9   Widget build(BuildContext context) {
10     return DefaultTabController(
11       length: 3,
12       child: Scaffold(
13         appBar: AppBar(
14           backgroundColor: Colors.limeAccent,
15           title: Text('Tab'),
16           centerTitle: true,
17           bottom: TabBar(tabs: [
18             Tab(
19               icon: Icon(Icons.chat),
20               text: "Chats",
21             ), // Tab
22             Tab(
23               icon: Icon(Icons.video_call),
24               text: "calls",
25             ), // Tab
26             Tab(
27               icon: Icon(Icons.settings),
28               text: "settings",
29             ) // Tab
30           ]), // TabBar
31         ), // AppBar
32         body: TabBarView(
33           children: [
34             Center(child: Text("chat page")),
35             Center(child: Text("calls page")),
36             Center(child: Text("settings page")),
37           ]), // TabBarView // Scaffold // DefaultTabController

```



Navigation Drawer

- Side menu used for app navigation.
- Implemented using the **Drawer widget**.
- Useful when the app has **5 or more pages**.
- Can be opened via swipe or menu icon.



```

1 import 'package:flutter/material.dart';
2 class drawer extends StatelessWidget {
3   const drawer({super.key});
4   @override
5   Widget build(BuildContext context) {
6     return Scaffold(
7       appBar: AppBar(
8         title: Text("empetz"),
9       ), // AppBar
10      body: Divider(),
11      drawer: Drawer(
12        backgroundColor: Colors.blueGrey,
13        shadowColor: Colors.amber,
14        surfaceTintColor: Colors.deepOrange,
15        child: ListView(
16          children: [
17            DrawerHeader(
18              decoration: BoxDecoration(color: Colors.brown),
19              child: Column(
20                children: [
21                  Text('RAHUL M G', style: TextStyle(fontSize: 20)),
22                  CircleAvatar(
23                    backgroundColor: Colors.blueAccent,
24                    radius: 40,
25                    child: Icon(Icons.person, size: 50, color: Colors.black), // ci
26                  ], // Column
27                ), // DrawerHeader
28                ListTile(
29                  leading: Icon(Icons.home),
30                  title: Text('Home'),
31                  onTap: () => Navigator.pop(context),
32                ), // ListTile
33                Divider(),
34                ListTile(
35                  leading: Icon(Icons.settings),
36                  title: Text('Settings'),
37                ), // ListTile // ListView // Drawer // Scaffold

```

RAHUL M G



🏠 Home

⚙️ Settings

Bottom Sheet

- Slides up from the **bottom of the screen**.
- Displays extra options or information.
- Does not block the main screen completely.



```

1 import 'package:flutter/material.dart';
2 class Bsheet extends StatefulWidget {
3   const Bsheet({super.key});
4   @override
5   State<Bsheet> createState() => _BsheetState();
6 }
7 class _BsheetState extends State<Bsheet> {
8   @override
9   Widget build(BuildContext context) {
10     return Center(
11       child: ElevatedButton(
12         child: Text('Show'),
13         onPressed: () {
14           showModalBottomSheet(
15             context: context,
16             builder: (BuildContext) {
17               return Container(
18                 height: 200,
19                 color: Colors.blue,
20                 child: Center(
21                   child: Column(
22                     mainAxisAlignment: MainAxisAlignment.center,
23                     mainAxisAlignment: MainAxisAlignment.min,
24                     children: [
25                       Text('Model'),
26                       ElevatedButton(onPressed: () => Navigator.pop(context),
27                         child: Text('close')) // ElevatedButton
28                     ],
29                   ), // Column
30                 ), // Center
31               ); // Container
32             });
33         ), // ElevatedButton
34       ); // Center
35     );
36   }
37 }

```

Show

Model

close

Custom Widgets

- User-defined reusable UI components.
- Benefits:
 - Code reusability
 - Better maintainability
 - Consistent UI design
- Used when:
 - Built-in widgets are insufficient
 - Same UI is reused in many places
 - Custom look or behavior is required



Complex Layouts and Responsive Design

- Design layouts for **different screen sizes**.
- Use appropriate **layout managers**.
- Follow **responsive design principles**.

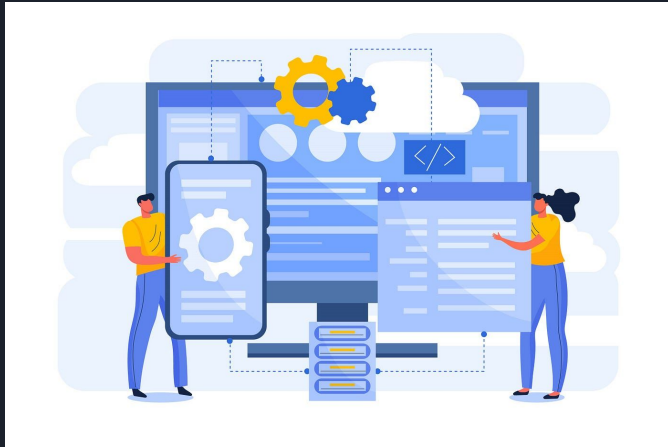


Advanced Animations and Motion Design

- Enhance UI using animations.
- Apply **motion design principles**.
- Use animation libraries like **Lottie**.

State Management at Scale

- Efficiently manage UI state in **large applications**.
- Use state management libraries.
- Handle performance and complexity issues.



Integrating UI with Backend Services

- Connect UI with backend APIs.
- Implement **data fetching & caching**.
- Handle **network errors gracefully**.

Exercise 1

Tabbar, Bottom sheet

```
lib > tabs.dart > _tabsState > build
1  import 'package:exercise_1/chat.dart';
2  import 'package:flutter/material.dart';
3  class tabs extends StatefulWidget {
4    const tabs({super.key});
5    @override
6    State<tabs> createState() => _tabsState();
7  }
8  class _tabsState extends State<tabs> {
9    @override
10   widget build(BuildContext context) {
11     return DefaultTabController(
12       length: 3,
13       child: Scaffold(
14         appBar: AppBar(
15           backgroundColor: Colors.yellowAccent,
16           title: Text('Bottom Sheet Sample'),
17           bottom: TabBar(
18             tabs: [
19               Tab(icon: Icon(Icons.message), text: 'Chats',),
20               Tab(icon: Icon(Icons.phone), text: 'Calls',),
21               Tab(icon: Icon(Icons.settings), text: 'Settings',)
22             ],
23           ), // TabBar
24         ), // AppBar
25         body: TabBarView(
26           children: [
27             Bottomsheet(),
28             Center(child: Text('Calls tab'),),
29             Center(child: Text('Settings Tab'),),
30           ],
31         ), // TabBarView
32       ), // Scaffold
33     ); // DefaultTabController
34   }
35 }
```

Bottom Sheet Sample



Chats



Calls



Settings

Calls tab

```

1 import 'package:flutter/material.dart';
2 class Bottomsheet extends StatefulWidget {
3   const Bottomsheet({super.key});
4   @override
5   State<Bottomsheet> createState() => _BottomsheetState();
6 }
7 class _BottomsheetState extends State<Bottomsheet> {
8   @override
9   Widget build(BuildContext context) {
10    return Center(
11      child: ElevatedButton(
12        child: Text('showModelBottomSheet'),
13        onPressed: (){
14          showBottomSheet(
15            context: context,
16            builder: (BuildContext){
17              return Container(
18                height: 200,
19                color: Colors.yellow,
20                child: Center(
21                  child: Column(
22                    mainAxisAlignment: MainAxisAlignment.center,
23                    children: [
24                      Text('Model Bottom Sheet'),
25                      ElevatedButton(
26                        onPressed: (){
27                          Navigator.pop(context);
28                        },
29                        child: Text('Close BottomSheet') // ElevatedButton
30                      ],
31                    ), // Column
32                  ), // Center
33                ); // Container
34              });
35            ), // ElevatedButton
36          ); // Center
37    }

```

← Bottom Sheet Sample



Chats



Calls



Settings

showModelBottomSheet

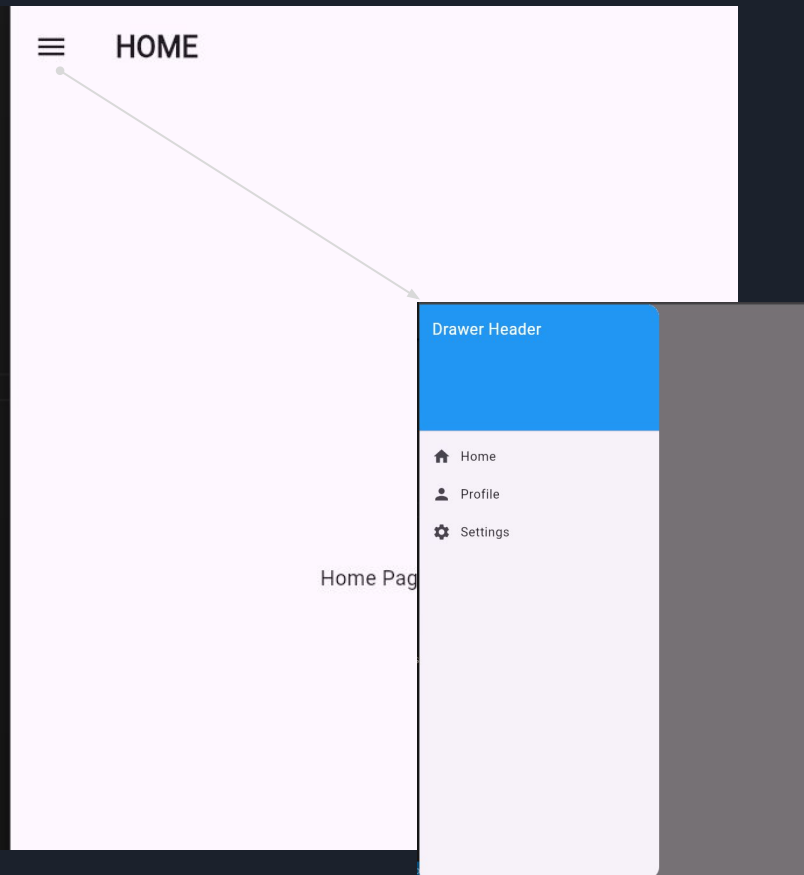
Model Bottom Sheet

Close BottomSheet

Exercise 2

Drawer,ListTile,Navigation

```
1 import 'package:exercise_2/prof.dart';
2 import 'package:exercise_2/settings.dart';
3 import 'package:flutter/material.dart';
4 class Drawerr extends StatefulWidget {
5   const Drawerr({super.key});
6   @override
7   State<Drawerr> createState() => _DrawerrState();
8   class _DrawerrState extends State<Drawerr> {
9     @override
10    Widget build(BuildContext context) {
11      return Scaffold(
12        appBar: AppBar(
13          title: Text('HOME',
14            style: TextStyle(fontSize: 20,fontWeight: FontWeight.bold),
15          ), // Text
16        ), // AppBar
17        drawer: Drawer(
18          child: ListView(
19            children: [
20              DrawerHeader(
21                decoration: BoxDecoration(color: Colors.blue),
22                child: Column(
23                  crossAxisAlignment: CrossAxisAlignment.start,
24                  children: [
25                    Text('Drawer Header',
26                      style: TextStyle(fontSize: 20,color: Colors.white),
27                    ), // Text
28                  ],
29                ), // Column
30              ), // DrawerHeader
```



```

30     ), // DrawerHeader
31     ListTile(
32       leading: Icon(Icons.home),
33       title: Text('Home'),
34       onTap: () => Navigator.pop(context),
35     ), // ListTile
36     ListTile(
37       leading: Icon(Icons.person),
38       title: Text('Profile'),
39       onTap: () {
40         Navigator.push(context, MaterialPageRoute(builder: (context) => prof()));
41       },
42     ), // ListTile
43     ListTile(
44       leading: Icon(Icons.settings),
45       title: Text('Settings'),
46       onTap: () {
47         Navigator.push(context, MaterialPageRoute(builder: (context) => sett()));
48       },
49     ), // ListTile
50   ],
51 ), // ListView
52 ), // Drawer
53 body: Center(
54   child: Text('Home Page'),
55 ), // Center
56 ); // Scaffold
57 }
58 }

```

Drawer Header

-  Home
-  Profile
-  Settings

Drawer Header

🏠 Home

👤 Profile

⚙️ Settings

```
1 import 'package:flutter/material.dart';
2 class prof extends StatefulWidget {
3   const prof({super.key});
4   @override
5   State<prof> createState() => _profState();
6 }
7 class _profState extends State<prof> {
8   @override
9   Widget build(BuildContext context) {
10    return Scaffold(
11      appBar: AppBar(
12        title: Text('Profile'),
13      ), // AppBar
14      body: Center(
15        child: Text('Profile Page'),
16      ), // Center
17    ); // Scaffold
18  }
19 }
20
21 import 'package:flutter/material.dart';
22 class sett extends StatefulWidget {
23   const sett({super.key});
24   @override
25   State<sett> createState() => _settState();
26 }
27 class _settState extends State<sett> {
28   @override
29   Widget build(BuildContext context) {
30    return Scaffold(
31      appBar: AppBar(
32        title: Text('Settings'),
33      ), // AppBar
34      body: Center(
35        child: Text('Settings Page'),
36      ), // Center
37    ); // Scaffold
38  }
39 }
```

← Profile

Profile Page

← Settings

Settings Page

Exercise 3

End Drawer, listview builder

```
1 import 'package:exercise_3/acc.dart';
2 import 'package:exercise_3/help.dart';
3 import 'package:exercise_3/sett.dart';
4 import 'package:flutter/material.dart';
5 class enddrawer extends StatefulWidget {
6   const enddrawer({super.key});
7   @override
8   State<enddrawer> createState() => _enddrawerState();
9 }
10 class _enddrawerState extends State<enddrawer> {
11   @override
12   Widget build(BuildContext context) {
13     return Scaffold(
14       appBar: AppBar(
15         title: Text('ListView With End Drwer', style: TextStyle(fontSize: 30)),
16         backgroundColor: Colors.grey,
17       ), // AppBar
18       endDrawer: Drawer(
19         child: ListView(
20           children: [
21             DrawerHeader(
22               decoration: BoxDecoration(color: Colors.blue),
23               child: Column(
24                 children: [
25                   Text('End Drawer', style: TextStyle(fontSize: 20, color: Colors.
26                 ],
27               ) // Column
28             ), // DrawerHeader
29           ],
30         ),
31       ),
32     );
33   }
```

ListView With

Item 1

Item 2

Item 3

Item 4

Item 5

Item 6

Item 7

Item 8

Item 9

Item 10

End Drawer

 Account

 Settings

 Help

```

29   ListTile(
30     leading: Icon(Icons.person),
31     title: Text('Account'),
32     onTap: () {
33       Navigator.push(context, MaterialPageRoute(builder: (context)=>acc()));
34     }
35   ), // ListTile
36   ListTile(
37     leading: Icon(Icons.settings),
38     title: Text('Settings'),
39     onTap: () {
40       Navigator.push(context, MaterialPageRoute(builder: (context)=>set()));
41     },
42   ), // ListTile
43   ListTile(
44     leading: Icon(Icons.help),
45     title: Text('Help'),
46     onTap: () {
47       Navigator.push(context, MaterialPageRoute(builder: (context)=>help()));
48     },
49   ), // ListTile
50 ],
51 ), // ListView
52 ), // Drawer
53 body: ListView.builder(
54   itemCount: 20,
55   itemBuilder: (BuildContext context,int index)
56   {
57     return ListTile(
58       title: Text('Item ${index +1}'),
59     ); // ListTile
60   }
61 ), // ListView.builder
62 ); // Scaffold

```

ListView With End Drwer

Item 1

Item 2

Item 3

Item 4

Item 5

Item 6

Item 7

Item 8

Item 9

Item 10

Item 11

Item 12

Item 13

Item 14

ListView With

End Drawer

Item 1

Item 2

Item 3

Item 4

Item 5

Item 6

Item 7

Item 8

Item 9

Item 10

 Account

 Settings

 Help

```
lib > sett.dart > _settState
1 import 'package:flutter/material.dart';
2 class sett extends StatefulWidget {
3   const sett({super.key});
4   @override
5   State<sett> createState() => _settState;
6 }
7 class _settState extends State<sett> {
8   @override
9   Widget build(BuildContext context) {
10    return Scaffold(
11      appBar: AppBar(
12        title: Text('Settings'),
13      ), // AppBar
14      body: Center(
15        child: Text('Settings Page'),
16      ), // Center
17    ); // Scaffold
18  }
19 }
```

```
lib > help.dart > ...
1 import 'package:flutter/material.dart';
2 class help extends StatefulWidget {
3   const help({super.key});
4   @override
5   State<help> createState() => _helpState;
6 }
7 class _helpState extends State<help> {
8   @override
9   Widget build(BuildContext context) {
10    return Scaffold(
11      appBar: AppBar(
12        title: Text('Help'),
13      ), // AppBar
14      body: Center(
15        child: Text('Help Page'),
16      ), // Center
17    ); // Scaffold
18  }
19 }
```

```
lib > acc.dart > ...
1 import 'package:flutter/material.dart';
2 class acc extends StatefulWidget {
3   const acc({super.key});
4   @override
5   State<acc> createState() => _accState;
6 }
7 class _accState extends State<acc> {
8   @override
9   Widget build(BuildContext context) {
10    return Scaffold(
11      appBar: AppBar(
12        title: Text('Account'),
13      ), // AppBar
14      body: Center(
15        child: Text('Account Page'),
16      ), // Center
17    ); // Scaffold
18  }
19 }
```

← Settings

Settings Page

← Help

Help Page

← Account

Account Page

Activity 3

IOS calculator

```
lib > calculator > CalculatorState > build
1 import 'package:flutter/material.dart';
2 class Calculator extends StatefulWidget {
3   const Calculator({super.key});
4   @override
5   State<Calculator> createState() => CalculatorState();
6 }
7 class CalculatorState extends State<Calculator> {
8
9   TextEditingController screen = TextEditingController();
10  TextEditingController screenone =TextEditingController();
11
12  String firstNumber = "";
13  String operator = "";
14  bool shouldResetScreen =false;
15
16  void addNumber(String num) {
17    setState(() {
18      String current = screen.text;
19      if (shouldResetScreen) {
20        screen.text = num;
21        shouldResetScreen = false;
22        return;
23      }
24      if (current == "0") {
25        if (num == "0") {
26          return;
27        } else {
28          screen.text = num;
29          return;
30        }
31      }
32      if (current.contains(".")) {
33        screen.text += num;
34        return;
35      }
36      screen.text += num;
37    });
38  }
39
40  void addPoint() {
41    setState(() {
42      String current = screen.text;
43
44      if (shouldResetScreen) {
45        screen.text = "0.";
46        shouldResetScreen = false;
47        return;
48      }
49
50      if (!current.contains(".")) {
51        screen.text += ".";
52      }
53    });
54  }
55 }
```

```
27
28 void clearAll(){
29   setState() {
30     screen.text="";
31     screenone.text="";
32     firstNumber="";
33     operator="";
34     shouldResetScreen=false;
35   });
36 }
```

```
38 void toggleSign(){
39   setState() {
40     double value =double.tryParse(screen.text) ?? 0;
41     if (value !=0){
42       value =value * -1;
43       if(value % 1 == 0){
44         screen.text=value.toInt().toString();
45       }else{
46         screen.text=value.toString();
47       }
48     }
49   });
50 }
51
52 void Operator(String op) {
53   setState() {
54     firstNumber = screen.text;
55     operator = op;
56     screenone.text="$firstNumber $operator";
57     shouldResetScreen=true;
58   });
59 }
60
61 void calculate() {
62   double num1 = double.tryParse(firstNumber) ?? 0;
63   double num2 = double.tryParse(screen.text) ?? 0;
64   double result = 0;
65   if (operator == "+") {
66     result = num1 + num2;
67   }
68   else if(operator == "-"){
69     result = num1 - num2;
70   }
71   else if(operator =="X"){
72     result =num1 * num2;
73 }
```

```
73
74   else if(operator =="/"){
75     if(num2 !=0){
76       result =num1 / num2;
77     }
78   } else{
79     result=0;
80   }
81 }
82   else if(operator =="%"){
83     result =num1 /100;
84   }
85   setState() {
86     if(operator=="%"){
87       screenone.text = "$firstNumber $operator";
88       if(result % 1==0){
89         screen.text =result.toInt().toString();
90       }
91     } else{
92       screen.text=result.toString();
93     }
94   }
95   else{
96     screenone.text = "$firstNumber $operator ${screen.text} ";
97     if(result % 1==0){
98       screen.text =result.toInt().toString();
99     }
100    else{
101      screen.text=result.toString();
102    }
103  }
104  shouldResetScreen =true;
105  });
106 }
```

```

108 Widget build(BuildContext context) {
109   return Scaffold(
110     backgroundColor: Colors.black,
111     body: Padding(padding: EdgeInsetsGeometry.all(10),
112       child: Column(
113         children: [
114           SizedBox(height: 30),
115           Padding(padding: EdgeInsetsGeometry.symmetric(horizontal: 50),
116             child: TextField(
117               controller: screenone,
118               readOnly: true,
119               textAlign: TextAlign.right,
120               style: const TextStyle(
121                 color: Colors.grey,
122                 fontSize: 30.0,
123                 fontWeight: FontWeight.bold), // TextStyle
124               decoration: const InputDecoration(
125                 border: InputBorder.none,
126                 contentPadding: EdgeInsets.all(5.0),
127               ),), // InputDecoration // TextField // Padding
128           SizedBox(height: 5),
129           Padding(padding: EdgeInsetsGeometry.symmetric(horizontal: 50),
130             child: TextField(
131               controller: screen,
132               readOnly: true,
133               textAlign: TextAlign.right,
134               style: const TextStyle(
135                 color: Colors.white,fontSize: 80.0,
136                 fontWeight: FontWeight.bold), // TextStyle
137               decoration: const InputDecoration(
138                 hintText: '0',
139                 hintStyle: TextStyle(color: Colors.white),
140                 border: InputBorder.none,
141                 contentPadding: EdgeInsets.all(5.0),
142               ),), // InputDecoration // TextField // Padding
143           SizedBox(height: 20),
144           Row(
145             mainAxisAlignment: MainAxisAlignment.spaceEvenly,
146             children: [
147               GestureDetector(
148                 onTap: clearAll,
149                 child: CircleAvatar(
150                   radius: 40,
151                   backgroundColor: Colors.grey,
152                   child: Center(
153                     child: Text("AC",
154                       style: TextStyle(
155                         fontSize: 30,color: Colors.black
156                       ),),), // TextStyle // Text // Center // Circle
157               GestureDetector(
158                 onTap: toggleSign,
159                 child: CircleAvatar(
160                   radius: 40,
161                   backgroundColor: Colors.grey,
162                   child: Center(
163                     child: Text("+/-",
164                       style: TextStyle(
165                         fontSize: 30,color: Colors.black
166                       ),),), // TextStyle // Text // Center // Circle
167               GestureDetector(
168                 onTap: () => Operator('%'),
169                 child: CircleAvatar(
170                   radius: 40,
171                   backgroundColor: Colors.grey,
172                   child: Center(
173                     child: Text("%",style: TextStyle(fontSize: 30,color: [
174                       ),), // Center // CircleAvatar // GestureDetector
175               GestureDetector(
176                 onTap: () => Operator("/"),
177                 child: CircleAvatar(
178                   radius: 40,
179                   backgroundColor: Colors.orangeAccent,
180                   child: Center(
181                     child: Text("÷",style: TextStyle(fontSize: 30,color: Colors.white)),
182                   ),), // Center // CircleAvatar // GestureDetector
183             ],), // Row
184           SizedBox(height: 20),
185           Row(
186             mainAxisAlignment: MainAxisAlignment.spaceEvenly,
187             children: [
188               GestureDetector(
189                 onTap: () => addNumber("7"),
190                 child: CircleAvatar(
191                   radius: 40,
192                   backgroundColor: const Color.fromARGB(255, 46, 46, 46),
193                   child: Center(
194                     child: Text("7",style: TextStyle(fontSize: 30,color: Colors.white)),
195                   ),), // Center // CircleAvatar // GestureDetector
196               GestureDetector(
197                 onTap: () => addNumber("8"),
198                 child: CircleAvatar(
199                   radius: 40,
200                   backgroundColor: const Color.fromARGB(255, 46, 46, 46),
201                   child: Center(
202                     child: Text("8",style: TextStyle(fontSize: 30,color: Colors.white)),
203                   ),), // Center // CircleAvatar // GestureDetector
204               GestureDetector(
205                 onTap: () => addNumber("9"),
206                 child: CircleAvatar(
207                   radius: 40,
208                   backgroundColor: const Color.fromARGB(255, 46, 46, 46),
209                   child: Center(
210                     child: Text("9",style: TextStyle(fontSize: 30,color: Colors.white)),

```

```

211      ),),), // Center // CircleAvatar // GestureDetector
212      GestureDetector(
213        onTap: () => Operator("x"),
214        child: CircleAvatar(
215          radius: 40,
216          backgroundColor: Colors.orangeAccent,
217          child: Center(
218            child: Text("x",style: TextStyle(fontSize: 30,color: Colors.white)),
219          ),),), // Center // CircleAvatar // GestureDetector
220      ),), // Row
221      SizedBox(height: 20),
222      Row(
223        mainAxisAlignment: MainAxisAlignment.spaceEvenly,
224        children: [
225          GestureDetector(
226            onTap: () => addNumber("4"),
227            child: CircleAvatar(
228              radius: 40,
229              backgroundColor: const Color.fromARGB(255, 46, 46, 46),
230              child: Center(
231                child: Text("4",style: TextStyle(fontSize: 30,color: Colors.white)),
232              ),),), // Center // CircleAvatar // GestureDetector
233          GestureDetector(
234            onTap: () => addNumber("5"),
235            child: CircleAvatar(
236              radius: 40,
237              backgroundColor: const Color.fromARGB(255, 46, 46, 46),
238              child: Center(
239                child: Text("5",style: TextStyle(fontSize: 30,color: Colors.white)),
240              ),),), // Center // CircleAvatar // GestureDetector
241          GestureDetector(
242            onTap: () => addNumber("6"),
243            child: CircleAvatar(
244              radius: 40,

```

```

245          backgroundColor: const Color.fromARGB(255, 46, 46, 46),
246          child: Center(
247            child: Text("6",style: TextStyle(fontSize: 30,color: Colors.white)),
248          ),),), // Center // CircleAvatar // GestureDetector
249          GestureDetector(
250            onTap: () => Operator("-"),
251            child: CircleAvatar(
252              radius: 40,
253              backgroundColor: Colors.orangeAccent,
254              child: Center(
255                child: Text("-",style: TextStyle(fontSize: 30,color: Colors.white)),
256              ),),), // Center // CircleAvatar // GestureDetector
257      ),), // Row
258      SizedBox(height: 20),
259      Row(
260        mainAxisAlignment: MainAxisAlignment.spaceEvenly,
261        children: [
262          GestureDetector(
263            onTap: () => addNumber("1"),
264            child: CircleAvatar(
265              radius: 40,
266              backgroundColor: const Color.fromARGB(255, 46, 46, 46),
267              child: Center(
268                child: Text("1",style: TextStyle(fontSize: 30,color: Colors.white)),
269              ),),), // Center // CircleAvatar // GestureDetector
270          GestureDetector(
271            onTap: () => addNumber("2"),
272            child: CircleAvatar(
273              radius: 40,
274              backgroundColor: const Color.fromARGB(255, 46, 46, 46),
275              child: Center(
276                child: Text("2",style: TextStyle(fontSize: 30,color: Colors.white)),
277              ),),), // Center // CircleAvatar // GestureDetector

```

```

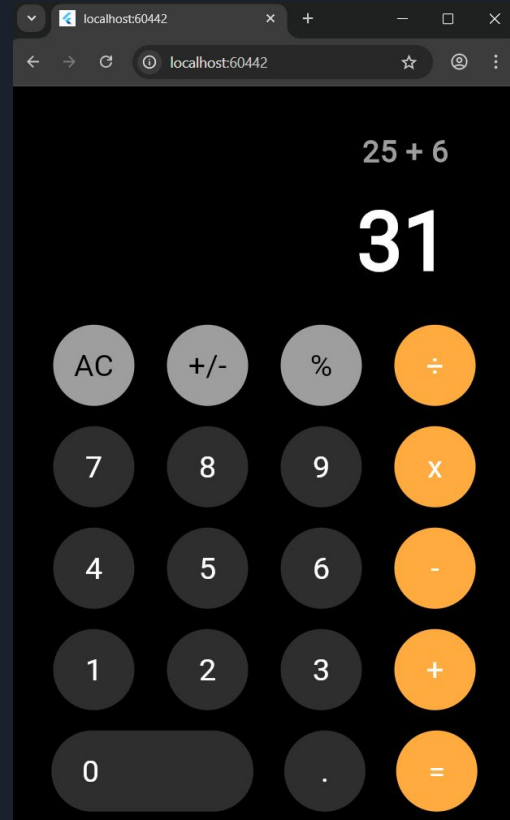
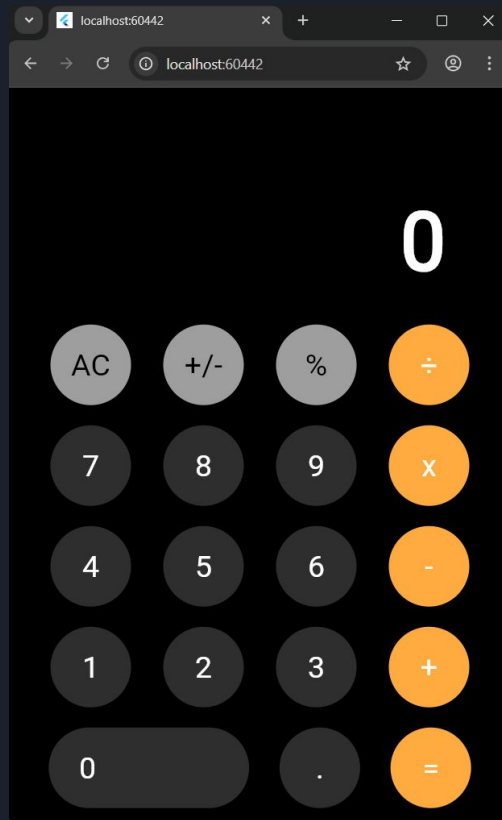
278          GestureDetector(
279            onTap: () => addNumber("3"),
280            child: CircleAvatar(
281              radius: 40,
282              backgroundColor: const Color.fromARGB(255, 46, 46, 46),
283              child: Center(
284                child: Text("3",style: TextStyle(fontSize: 30,color: Colors.white)),
285              ),),), // Center // CircleAvatar // GestureDetector
286          GestureDetector(
287            onTap: () => Operator("+"),
288            child: CircleAvatar(
289              radius: 40,
290              backgroundColor: Colors.orangeAccent,
291              child: Center(
292                child: Text("+",style: TextStyle(fontSize: 30,color: Colors.white)),
293              ),),), // Center // CircleAvatar // GestureDetector
294      ],), // Row
295      SizedBox(height: 20),
296      Row(
297        mainAxisAlignment: MainAxisAlignment.spaceEvenly,
298        children: [
299          GestureDetector(
300            onTap: () => addNumber("0"),
301            child: Container(
302              width: 190,
303              height: 80,
304              decoration: BoxDecoration(
305                borderRadius: BorderRadius.circular(40),
306                color: Color.fromARGB(255, 46, 46, 46),
307              ), // BoxDecoration
308              child: Center(
309                child: Row(
310                  children: [

```

```

310     children: [
311       SizedBox(width: 30),
312       Text("0", style: TextStyle(fontSize: 30, color: Colors.white)),
313     ],
314   ),),), // Row // Center // Container // GestureDetector
315   GestureDetector(
316     onTap: () => addNumber("."),
317     child: CircleAvatar(
318       radius: 40,
319       backgroundColor: const Color.fromARGB(255, 46, 46, 46),
320       child: Center(
321         child: Text(".", style: TextStyle(fontSize: 30, color: Colors.white)),
322       ),), // Center // CircleAvatar // GestureDetector
323   GestureDetector(
324     onTap: () => calculate(),
325     child: CircleAvatar(
326       radius: 40,
327       backgroundColor: Colors.orangeAccent,
328       child: Center(
329         child: Text("=", style: TextStyle(fontSize: 30, color: Colors.white)),
330       ), // Center
331     ), // CircleAvatar
332   ), // GestureDetector
333 ], // Row
334 ], // Column
335 ), // Column
336 ), // Padding
337 ); // Scaffold
338 }
339 }
340 }

```



Conclusion

- Advanced UI techniques improve:
 - User experience
 - App scalability
 - Maintainability
- Continuous learning and practice are essential.